

Aarsh Sachdeva

Chicago, IL | aarsh98a@gmail.com | (312) 285-5318
LINKEDIN | **PL-300: Microsoft Certified Power BI Data Analyst** | GITHUB

Experience

Data Scientist

HCL Technologies Limited

Noida Sector-126, India

02/2022 – 07/2024

- **Digital Transformation:** Contributed to \$125K+ in annual cost savings (23% ROI improvement) by developing automation and data analytics solutions that analyzed customer service conversations, automated workflows, and generated business insights.
- **Machine Learning:** Developed a ML model to automatically classify 10M+ customer conversations using NLP techniques and XGBoost, improving classification performance by 38% (Macro F1: 0.60 → 0.82).
- **Data Visualization:** Built and deployed 15+ **Power BI** dashboards tracking customer sentiment, employee performance, and operational KPIs across regions and business units, enabling faster data-driven decision-making.
- **Data Engineering & SQL:** Engineered **SQL** data pipelines to extract, validate, transform, and prepare large-scale datasets from relational databases, establishing reliable data methodologies for analytics and machine learning workflows.
- **Predictive analytics:** Performed trend analysis on call volume using **time series forecasting** models, including ARIMA, SARIMA, and Prophet, to support operational planning and data driven decision-making.
- **NLP based Sentiment Analysis:** Applied NLP techniques to analyze customer service conversations and identify sentiment patterns across 10M+ customer interactions, generating insights for employee performance and customer satisfaction.
- **Process Automation:** Automated 25+ operational reporting workflows and reduced using manual effort by 40% using **VBA** and **Microsoft Power Automate**, improving process efficiency, quality control, reporting accuracy, and consistency.

Teaching Assistant

Coursera

Chicago, IL

08/2025 – 05/2026

- Mentored 250+ students over 2 semesters as a TA for **Machine Learning** and **Bayesian Statistics**, conducting twice-weekly office hours, providing one-on-one academic support, assisting with professor-led live sessions, and maintaining grade records in **Excel**.
- Recognized as the best teaching assistant among 20+ TAs for outstanding teaching and contributions to the Mathematics department.

Projects

Variance Reduced Pricing for Path dependent Options (Monte Carlo & Option Pricing) ↗

- Engineered a variance-reduced Monte Carlo engine for path dependent option pricing, achieving 10-21% SE reduction over Vanilla MC via Control Variates and Antithetic Variates.
- Implemented Quasi-Monte Carlo via Sobol Sampling, achieving 100x SE reduction and 55x cost reduction at equivalent simulation budget over Vanilla MC.

Bike Sharing Demand Analysis (GLM Regression & Time Series) ↗

- Estimated demand regression models using GLMs (Poisson, Negative Binomial) across temporal variables including time of week effects, achieving deviance-per-degree-of-freedom of 0.87 with Negative Binomial model.
- Validated model robustness through overdispersion testing, residual diagnostics and ad-hoc statistical calculations, benchmarking against ML alternatives for predictive reliability.

Education

Illinois Institute of Technology, Chicago, IL

Master of Science in Applied Mathematics (GPA - 3.6/4)

08/2024 – 05/2026

- **Coursework:** Linear Algebra, Time Series Analysis, Bayesian Statistics, Regression, Monte Carlo Methods

JSSATE, India

Bachelor of Technology in Electronics and Communication

08/2017 – 05/2021

Skills and Certifications

Programming & Databases: Python, R, SQL, Excel (Advanced), VBA, AWS Lamda, AWS Sagemaker

Machine Learning & Statistics: Predictive Modeling, A/B Testing, Time Series Forecasting, Regression, Hypothesis Testing, Experimental Design, Monte Carlo Simulation, Generalized Linear Models (GLM)

Data Science & Analytics: Pandas, NumPy, Scikit-learn, Statsmodels, SciPy, Feature Engineering, Model Validation, ETL Pipelines

Tools & Data Visualization: MLflow, Git, Power BI, Tableau, Matplotlib